

Math 241, F18
Quiz 1

Name: Solution

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz. Problem

1. (10 points) Given two three-dimensional vectors $\mathbf{u} = 3\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$ and $\mathbf{v} = \langle 4, 0, 3 \rangle$.
(a) Compute $3\mathbf{u} + 2\mathbf{v}$ and $\mathbf{u} - \mathbf{v}$.

$$\begin{aligned} 3\mathbf{u} + 2\mathbf{v} &= 3\langle 3, -2, 6 \rangle + 2\langle 4, 0, 3 \rangle \\ &= \langle 9, -6, 18 \rangle + \langle 8, 0, 6 \rangle = \langle 17, -6, 24 \rangle \\ \mathbf{u} - \mathbf{v} &= \langle 3, -2, 6 \rangle - \langle 4, 0, 3 \rangle = \langle -1, -2, 3 \rangle \end{aligned}$$

- (b) Find the lengths $|\mathbf{u}|$ and $|\mathbf{v}|$.

$$\begin{aligned} |\mathbf{u}| &= \sqrt{3^2 + (-2)^2 + 6^2} = \sqrt{9 + 4 + 36} = \sqrt{49} = 7 \\ |\mathbf{v}| &= \sqrt{4^2 + 0^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5 \end{aligned}$$

- (c) Find a unit vector that has the same direction as $\mathbf{u}$.

$$\frac{\mathbf{u}}{|\mathbf{u}|} = \frac{3\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}}{7} = \frac{3}{7}\mathbf{i} - \frac{2}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}$$

- (d) Find the angle between the two vectors $\mathbf{u}$ and $\mathbf{v}$.

$$\begin{aligned} \mathbf{u} \cdot \mathbf{v} &= \langle 3, -2, 6 \rangle \cdot \langle 4, 0, 3 \rangle \\ &= 3 \cdot 4 + (-2) \cdot 0 + 6 \cdot 3 = 30 \\ \theta &= \cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| |\mathbf{v}|} \right) = \cos^{-1} \left(\frac{30}{7 \cdot 5} \right) \\ &= \cos^{-1} \left(\frac{6}{7} \right) \end{aligned}$$